

Operations & Logistics Intelligence Project

Supply Chain Graduation Project (2022-2024)

Instructor

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SUPPLY CHAIN
LOGISTICS

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1. Project Overview

This project involves a comprehensive analysis of a realistic logistics operational database spanning three years (2022-2024).

360,000+

Raw Records

14

Interconnected Tables

Strategic Goals & Objectives



Optimize Fleet Performance



Reduce Operational Costs



Enhance Service Reliability

Questions to Explore

01 Which routes generate the highest profit margin after fuel costs?

02 How does driver tenure correlate with fuel efficiency and on-time performance?

03 What's the optimal preventive maintenance interval to minimize total cost of ownership?

04 Which customers have the highest revenue-per-load and best payment terms?

05 How do seasonal patterns affect equipment utilization and revenue?

06 What safety incident patterns exist by driver experience level?

07 Which city pairs have the most reliable on-time delivery performance?

08 How does truck age impact maintenance costs and downtime?

Data Pruning Process

We cleaned the **trips** table by filtering out missing values in key identifier fields (**driver_id**, **truck_id**, **trailer_id**). This ensures statistical integrity and unbiased profitability metrics for our analysis.

BEFORE: Raw Data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 85410 entries, 0 to 85409
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   trip_id                85410 non-null  object
1   load_id                85410 non-null  object
2   driver_id              83696 non-null  object
3   truck_id               83738 non-null  object
4   trailer_id             83730 non-null  object
5   dispatch_date          85410 non-null  object
6   actual_distance_miles  85410 non-null  int64
7   actual_duration_hours  85410 non-null  float64
8   fuel_gallons_used      85410 non-null  float64
9   average_mpg            85410 non-null  float64
10  idle_time_hours        85410 non-null  float64
11  trip_status            85410 non-null  object
dtypes: float64(4), int64(1), object(7)
memory usage: 7.8+ MB
```



AFTER: Cleaned Dataset

```
Index: 80458 entries, 0 to 85409
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   trip_id                80458 non-null  object
1   load_id                80458 non-null  object
2   driver_id              80458 non-null  object
3   truck_id               80458 non-null  object
4   trailer_id             80458 non-null  object
5   dispatch_date          80458 non-null  datetime64[ns]
6   actual_distance_miles  80458 non-null  int64
7   actual_duration_hours  80458 non-null  float64
8   fuel_gallons_used      80458 non-null  float64
9   average_mpg            80458 non-null  float64
10  idle_time_hours        80458 non-null  float64
11  trip_status            80458 non-null  object
dtypes: datetime64[ns](1), float64(4), int64(1), object(6)
memory usage: 8.0+ MB
```

1 - Trips Table: Data cleaning

We performed **Data filtering** on the **trips** table to remove inconsistent records.

By filtering out **incomplete data**, we ensured the **Statistical Integrity** of our performance metrics, preventing biased results in profitability analysis.

We removed incomplete rows, resulting in **80,458 rows** of high-quality data and converted date columns to a standard format.

Data Cleaning Summary

- **Total Raw Records:** 85,410
- **Records Removed:** 4,952
- **Data Loss Rate:** 5.80%
- **Final Dataset:** 80,458 rows

5.8% Gap Discovered in trip records.

Recommendation: Fix logging software to make identifier fields mandatory.

"In a real trucking company, you cannot have a trip without a driver or a truck. It's like a ghost trip".

This prevents data loss and ensures 100% accurate future reporting without manual cleaning.

🔍 فحص القيم المفقودة في جدول الرحلات (Trips):

```
trip_id      0
load_id      0
driver_id    1714
truck_id     1672
trailer_id    1680
dispatch_date 0
actual_distance_miles 0
actual_duration_hours 0
fuel_gallons_used 0
average_mpg   0
idle_time_hours 0
trip_status   0
dtype: int64
```

Data Synchronization: Maintaining Integrity Across Tables

After cleaning the **Trips** table, other tables (Fuel, Delivery Events, Safety, Loads) were synchronized to ensure analysis of valid trips only.

Why this is important?

- 1. Consistency:** Ensuring **trip_id** exists in all tables; no fuel costs for a deleted trip!
- 2. Accurate KPIs:** Keeping deleted records leads to errors in "Total Cost" and profit analysis.
- 3. Prevent Orphaned Records:** Prevents "ghost" data that doesn't belong to any real trip.



Clean Trip IDs were used as a filter for all linked tables.

How it works: If a trip ID is deleted from the main table, it is automatically removed from all child tables.

Comprehensive Data Cleaning & Synchronization

Drivers & Trips Synchronization

Driver records were cleaned and synchronized.

Number of drivers before filtering: **150**

Remaining drivers (who have trips): **124**

Fuel & Loads Refinement

- Fuel table cleaned & dates converted.
- Loads table texts cleaned; revenue calculated via load ID.

Fleet & Infrastructure Tables

The following specialized tables were thoroughly cleaned and verified:

- **Assets:** Truck table, Trailers
- **Operations:** Maintenance Records, Routes, Facilities
- **Safety:** Safety Incidents Profiling



All supporting operational tables have been fully cleaned and synchronized.

Categorizing Delivery Punctuality

Arrival Status Methodology

In the `Delivery Events` table, we identified negative time differences, which represent early arrivals.

We created a new column called `arrival_status` to categorize each event:

- **Early:** Negative time difference.
- **Late:** Positive time difference.

This transforms raw numbers into clear, visual insights about driver punctuality.

Data Transformation Example

	event_type	arrival_delay_minutes	arrival_status	on_time_flag
0	Pickup	178.931970	Late	False
1	Delivery	-100.562935	Early	True
2	Pickup	-22.556526	Early	True
3	Delivery	222.327164	Late	False
4	Pickup	-31.963839	Early	True

Note: The `on_time_flag` is derived directly from the `arrival_status` to simplify KPI reporting.

KPI Calculation Methodologies

OTD (On-Time Delivery %)

The primary service-level metric for customer satisfaction.

$$\text{OTD\%} = (\text{'Early' or 'On-Time'} / \text{Total}) \times 100$$

RPM (Revenue Per Mile)

Measures the actual earning power of each mile driven.

$$\text{RPM} = \text{Total Load Revenue} / \text{Total Distance Miles}$$

Asset Utilization Rate

Measures the percentage of time trucks are actively generating revenue vs. sitting idle.

$$\text{Avg. Util} = \text{Average of (util_rate)} \times 100$$

MPG (Miles Per Gallon)

Tracks fuel efficiency across different truck models and driver behaviors.

$$\text{MPG} = \text{Total Distance} / \text{Total Fuel Purchases}$$

These formulas are used to calculate the primary metrics displayed in the KPI dashboard.



Fleet Utilization Dashboard

Select
filter

home_t...

- ☐ Atlanta
- ☐ Charl...
- ☐ Colu...
- ☐ Dallas
- ☐ Denver
- ☐ Detroit
- ☐ Hous...
- ☐ India...

55K
Downtime

\$0.036
Maintenance Cost pe...

2.8M
Revenue per Truck

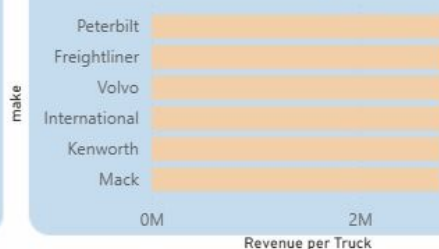
83.0%
Truck Utilization Rate

36K
Avg. Miles per ...

Maintenance Cost per Mile by Year



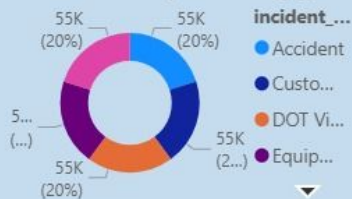
Revenue per Truck by make



Truck Utilization Rate by Year



Downtime by incident_type



Truck Utilization Rate by make



Total Maintenance Costs by make





Customer Performance Dashboard

Select
filter

Total
Revenue
by cust...

- ☐ Co...
- ☐ De...
- ☐ Spot

Total
Revenue
by prima...

- ☐ Auto...
- ☐ Cons...

\$262.5M

Total Revenue

200

Total customers

427

Loads per Customer

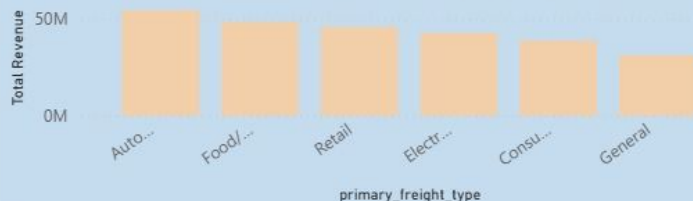
\$1.3M

Total Revenue per ...

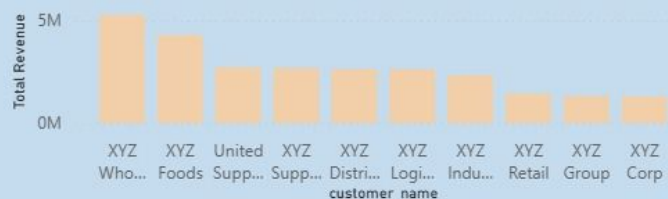
\$3.1K

Average Revenue per C...

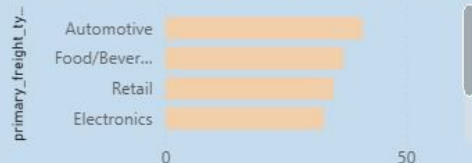
Total Revenue by primary_freight_type



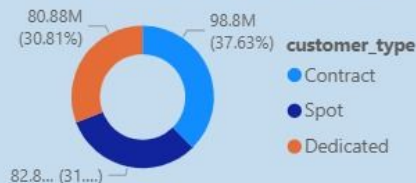
Top 10 Revenue by customer_name



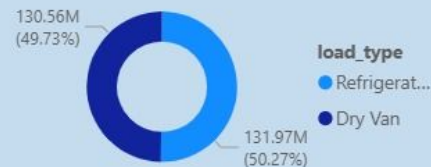
Total customers by primary_freight_type



Total Revenue by customer_type



Total Revenue by load_type



Logistics Performance Executive Report: Dashboard 1

1. Financial & Route Profitability

Metric: Rev (\$10.28M) vs. Fuel (\$91.83M)

Insight: Route Profit Margin (-845.34) shows severe deficit; fuel is 95% of OpEx.

Recommendation: Immediate Rate Renegotiation or Dynamic Fuel Surcharge.

2. Asset Utilization & Maintenance

Metric: 83% Utilization; 55,103 Hours Total Downtime

Insight: High activity but excessive shop time suggests inefficiency.

Recommendation: Shift to "Predictive Maintenance" using corrected vehicle age data.

3. Operational & Driver Efficiency

Metric: 7.01 Avg Idle Hours; 6.50 Avg MPG

Insight: Excessive idling at facilities (detention) undermines fuel efficiency.

Recommendation: Launch Zero-Idling program and optimize scheduling.

4. Safety & Risk Management

Metric: \$1,048,672 Total Incident Claims

Insight: Incidents >\$1M significantly impact net profit and the bottom line.

Recommendation: Deploy Safety Workshops in high-incident regions.

This report highlights critical areas for immediate operational and financial improvement.

Driver Performance Analysis Report

1. Fleet Productivity & Activity

Metric: Avg Miles/Driver: 26.81K | Drivers: 150

Insight: High workforce utilization is achieved, but pressures maintenance schedules.

Rec: Monitor HOS and distribute long-haul trips to prevent burnout.

2. Fuel Efficiency & Environment

Metric: Avg MPG: 6.50 | Avg Fuel Cost: \$612.2K

Insight: Marginal MPG improvements can save hundreds of thousands of dollars.

Rec: Launch "Eco-Driving" Training to target 7.0 MPG (7% reduction).

3. Operational Bottlenecks

Metric: Idle: 7.01 Hrs | Detention: 91.56 Min

Insight: Non-productive idling and waiting consume fuel and lower satisfaction.

Rec: Use Auto Engine Management and renegotiate detention terms.

4. Safety & Compliance

Metric: Accidents: 170 | Claims: \$1.05M

Insight: Accident rate exceeds 1 per driver; claims have crossed the \$1M mark.

Rec: Pivot to Safety Scorecard incentives to reduce claim liabilities.

This performance summary identifies critical areas for safety, efficiency, and productivity gains.

Cost Management Analysis Report

1. Fuel Expenditure Domination

Metric: Total Fuel Cost: **\$91.83M** | Avg/Unit: **\$3.90**

Insight: Primary expense; 1% reduction saves nearly **\$1M** annually.

Recommendation: Centralized Fuel Procurement & GPS-based Route Optimization.

2. Maintenance Efficiency

Metric: Total Maintenance Cost: **\$4.33M**

Insight: Well-managed at **4.7%** of fuel spend; cross-reference with availability.

Recommendation: Shift focus to Preventative Maintenance for older units.

3. Safety & Incident Liability

Metric: Incident Costs: **\$1M**

Insight: "Lost capital" eroding net profit margin directly.

Recommendation: Reinvest in ADAS & training; 25% reduction adds **\$250K** to bottom line.

Critical areas for financial and operational optimization across the fleet.

Shipment & Trip Performance Comprehensive Report

1. Fleet Financial Productivity

Metric: Avg Rev/Load: \$3.07K

Insight: Tight margins due to high fuel costs.

Rec: Review "Under-Performing" routes to push avg >\$3.5K.

2. Payload Capacity Utilization

Metric: Avg Load Weight: 27.48K lbs

Insight: Moderate capacity; significant "Empty Space" remains.

Rec: Implement Load Optimization for 35K-40K lbs range.

3. Operational Punctuality Gap

Metric: Avg Arrival Delay: 105.40 Mins

Insight: ~1h 45m delay drives detention costs and frustration.

Rec: Audit Dispatch Scheduling for better estimations.

4. Delivery Cycle Speed

Metric: Avg Delivery Time: 1.00 Day

Insight: High velocity in short-to-medium haul segment.

Rec: Maintain 1-day standard as a Competitive Advantage.

5. Service Reliability

Metric: Rejected Loads: 0 (Blank)

Insight: 100% Commitment Rate is a rare, valuable KPI.

Rec: Use "100% Reliability" as a lead marketing message.

6. Workforce Engagement

Metric: Total Number of Trips: (Insert Total)

Insight: High volume handled by 150 drivers daily.

Rec: Balance volume with 83% utilization to avoid wear.

Comprehensive analysis of financial, operational, and service metrics across the fleet.

Fleet Utilization Executive Report

1. Fleet Productivity & Capacity

Metric: Truck Utilization Rate: **0.83 (83%)**

Insight: Minimal idle assets, strong demand.

Rec: Maintain 17% buffer for spikes.

2. Mechanical Performance & Endurance

Metric: Avg Miles/Truck: **36.15K**

Insight: High stamina but accelerates wear.

Rec: Use data for Automated Maintenance Alerts.

3. Asset Revenue Generation

Metric: Revenue per Truck: **\$2.80M**

Insight: Strong ROI per vehicle.

Rec: Replace older units with lower revenue.

4. Maintenance Cost Efficiency

Metric: Cost per Mile: **\$0.04**

Insight: Exceptionally efficient routine repairs.

Rec: Benchmark against industry standards.

5. The Operational "Red Flag" (Downtime)

Metric: Total Downtime: **55K Hours**

Insight: Massive loss of potential revenue. Trucks sitting in the shop instead of on the road.

Recommendation: Reducing downtime by 10% (5,500 hours) unlocks revenue without new assets.

Operational excellence through proactive maintenance and downtime reduction.

Shipment & Trip Performance Analysis Report

1. Load Efficiency & Revenue

Metric: Rev/Load: \$3.07K | Avg Weight: 27.48K

Insight: Revenue flat despite weight correlating with truck capacity.

Rec: Explore Load Consolidation to maximize legal weight limits.

2. Delivery Timeliness & Reliability

Metric: Delay: 105.40 | Avg. Delivery Time: 1

Insight: 105 min delay is a significant operational failure.

Rec: Implement Predictive ETA; investigate dispatch scheduling.

3. Operational Integrity

Metric: Canceled/Rejected Loads: Zero

Insight: "Zero Rejection" signifies high Service Reliability.

Rec: Leverage reliability record as a key marketing selling point.

Analysis of performance metrics to drive load efficiency and operational excellence.

Strategic Operational Insights & Action Plan

1. Profitability Crisis

Insight: Total Fuel Costs (\$91.8M) are severely outpacing Route Revenue (\$10.3M).

Action: Implement **Dynamic Fuel Surcharges** and renegotiate high-distance rates.

2. Asset Inactivity

Insight: **55,103 Hours** of downtime stalling revenue despite **83%** utilization.

Action: Transition to **Predictive Maintenance** to minimize unscheduled shop time.

3. Operational Leaks

Insight: **Idling (7.0 hrs)** and **Detention (91.6 min)** drive fuel waste.

Action: Launch **"Zero-Idling" Incentives** and optimize scheduling.

4. Financial Risk

Insight: Safety claims have exceeded **\$1M**, eroding net profitability.

Action: Conduct **Regional Safety Workshops** to reduce claim costs.

Critical operational findings requiring immediate execution of the action plan.

The story to be continued..

Thank You